

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CE/IT)

May 2012

IT303 Advanced Database Management System

Date: 03.05.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are the two types of indices? Explain any one of them in detail. [04]
(b) What are persistent objects and transient objects? [03]
- Q - 2 (a) Explain why it may be impractical to require serializability for long-duration transactions. [06]
(b) State the two-phase commit protocol, and discuss the implications of a failure on the part of (i) the coordinator and (ii) a participant, during each of the two phases. [05]
(c) What is a view? How it is related to data independence? [03]

OR

- Q - 2 (a) What do you mean by background processes with respect of transaction server? Explain Database Writer background process and Log Writer (LGWR) background process. [06]
(b) State the difference between views and snapshot with example, in general what we are using in SQL? [05]
(c) What are the basic cursor attributes used in PL/SQL? [03]
- Q - 3 (a) Define Distributed DBMS. Explain its types of Transparency. [05]
(b) Give the differences between client server architecture and a centralized computing system. [05]
(c) Explain the concept of GRANT and REVOKE with example. [04]

OR

- Q - 3 (a) What is distributed transaction and how does it differ from remote transaction? Explain with proper example. [05]
(b) What do you mean by authentication? Differentiate between object privilege and system privilege. [05]
(c) Since indices speed query processing, why might they not be kept on several search keys? List as many reasons as possible. [04]

SECTION – II

- Q - 4 (a) Attempt the following: [04]
(i) Differentiate explicit and an implicit lock.
(ii) Explain how procedures are called in a PL/SQL block and what happen if we will not assign anything in OUT parameter type in a procedure?
- (b) Is a high-performance transaction system necessarily a real-time system? Why or why not? [03]
- Q - 5 (a) Explain the following terms: [08]
(i) Three-valued logic (3VL) (iii) Type inheritance
(ii) Instance (iv) Null values
- (b) What form of parallelism (interquery, interoperation, or intraoperation) is likely to be the most important for each of the following tasks? [06]
(i) Increasing the throughput of a system with many small queries.
(ii) Increasing the throughput of a system with a few large queries, when the number of disks and processor is large.

OR

- Q - 5 (a) Answer the followings. [08]
(i) Let the relation R contains following tuples.
(6, 5, 4)
(UNK, 5, 4)
(6, UNK, 4)
(UNK, UNK, UNK)
Find the truth value for the following given expression.
(a) $\text{FORALL } V (V.B > 1 \text{ OR IS_UNK}(V.B))$
(b) $\text{EXISTS } V (\text{MAYBE}(V.C > 3))$
- (ii) Compare RDBMS and ORDBMS. Describe an application scenario for which you would choose ORDBMS or RDBMS.
- (b) Describe the benefits and drawbacks of pipelined parallelism. [04]
- (c) How does the concept of an object in the object-oriented model differ from the concept of an entity in the entity-relationship model? [02]
- Q - 6 (a) Answer the following (Any Three) [12]
(1) What are stored-procedures and what are the advantages of using them?
(2) Differentiate open hashing and closed hashing.
(3) Like database systems, workflow systems also require concurrency and recovery management. List three reasons why we cannot simply apply a relational database system using 2PL, physical undo logging, and 2PC.
(4) Can a domain contain an UNK? Explain in detail.
(5) What are the major the differences between DBMS and RDBMS?
- (b) You are creating many users who require the same privileges. What should you use to make your job easier? Explain in detail. [02]
